

Jyotiraditya

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PROFESSIONAL SUMMARY

Machine Learning Engineer with early-career experience building end-to-end LLM pipelines and retrieval-augmented generation systems. Delivered a full-stack trading analytics platform that enabled real-time stock scoring, and optimized self-checkout modules, reducing checkout time by 15% in high-volume retail. Skilled in translating research prototypes into production-ready AI services on cloud infrastructure. Seeking to apply this blend of AI and full-stack expertise to accelerate intelligent product development.

EDUCATION

New Jersey Institute of Technology, YWCC

Master of Science, Computer Science

Aug 2024 - May 2026

Newark, NJ

Manipal Institute of Technology

Bachelor of Technology, Electrical and Electronics Engineering, Fundamentals of Computing

Jul 2018 - Jul 2022

Manipal, India

PROFESSIONAL EXPERIENCE

Build City

Software Engineer - Full Stack

New York City, NY · May 2025 - Aug 2025

- Built and deployed a full-stack trading analytics platform using FastAPI and the Financial Modeling Prep API, enabling real-time stock scoring and multi-ticker comparison, which helped users make faster investment decisions
- Developed a modular Next.js frontend with Tailwind CSS, deployed builds on Vercel, and optimized routing and performance, resulting in faster page loads and a responsive interface across multiple device form factors
- Implemented backend services on Render, debugging live production issues using server logs and API-level monitoring to ensure system reliability, availability, and operational stability.
- Designed PostgreSQL data models and RESTful API layers that supported extensible analytics workflows and scalable data ingestion, enabling rapid addition of new features and meeting evolving business needs.

Falabella

Frontend Software Developer

Bengaluru, India · Aug 2022 - Aug 2024

- Led development and optimization of self-checkout modules in React, reducing checkout time by ~15% across high-volume retail stores and peak customer traffic periods during daily production operations.
- Diagnosed and resolved over 90% of production issues via remote desktop sessions, consistently meeting SLAs in fast-paced, customer-facing retail environments with minimal service disruption.
- Implemented Cypress for component and end-to-end test automation, cutting manual QA effort by ~40% while improving regression coverage and release reliability across frequent deployment cycles.
- Directed multi-store rollouts across ~9 locations using GCP log analysis, proactively identifying failures and improving deployment reliability during large-scale production launches.

Software Development Intern

Jan 2022 - Jul 2022

- Improved data retrieval efficiency by 25% via RESTful APIs built with Node.js and PostgreSQL.
- Supported the e-commerce backend built with Node.js, TypeScript, and PostgreSQL, reliably processing ~1000 API requests each day and keeping the service available for customers

PROJECTS

Adaptive LLM-Based Relevance-Diversity Optimization Framework [top-k](#)

- Engineered an adaptive LLM-driven framework to identify top-k hotel groups by jointly modeling semantic relevance and inter-review diversity across candidate sets under constrained evaluation budgets and query limits.
- Designed a minimum-valid-certificate (MVC) guided querying loop to selectively acquire the most informative unknown relevance and diversity scores during iterative ranking refinement cycles.
- Formulated range-based bounds and dominance pruning logic to eliminate suboptimal candidates early and reduce model evaluations without sacrificing ranking correctness or decision stability.
- Achieved up to 65% reduction in total LLM evaluations while preserving ranking fidelity, interpretability, and selection consistency across diverse query scenarios and datasets

Financial Research Copilot [financial-research-copilot](#)

- Built a Financial Research Copilot using FastAPI, Next.js, LangChain, FAISS, and LLMs to analyze financial documents and generate actionable insights.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline with PDF ingestion, document chunking, embeddings, vector search, and source-grounded question answering.
- Developed AI-powered features including Executive Summary Generation, Financial Intelligence Reports, and Investment Memo Creation to automate equity research workflows.
- Designed an interactive research platform with multi-document management, conversational chat interface, persistent chat history, and real-time financial analysis for enhanced user experience and decision-making.

TECHNICAL SKILLS

- Languages:** Java, Python, JavaScript, TypeScript, SQL, HTML/CSS
- Frameworks & Libraries:** React.js, Next.js, Node.js, FastAPI, Flask, LangChain, Hugging Face Transformers, Cypress, PyTorch, Scikit-learn, TensorFlow, XGBoost
- Tools & Platforms:** Git, PostgreSQL, Google Cloud Platform (GCP), OpenAI API, Hugging Face Hub, Docker, Vercel, Render
- Core Competencies:** Full-Stack Development, RESTful API Design, Prompt Engineering, LLM Integration, Data Visualization, AI Agents & Conversational AI, Model Deployment & MLOps, Feature Engineering & Data Pipelines, Retrieval-Augmented Generation